

MT4013 - Material Sensor

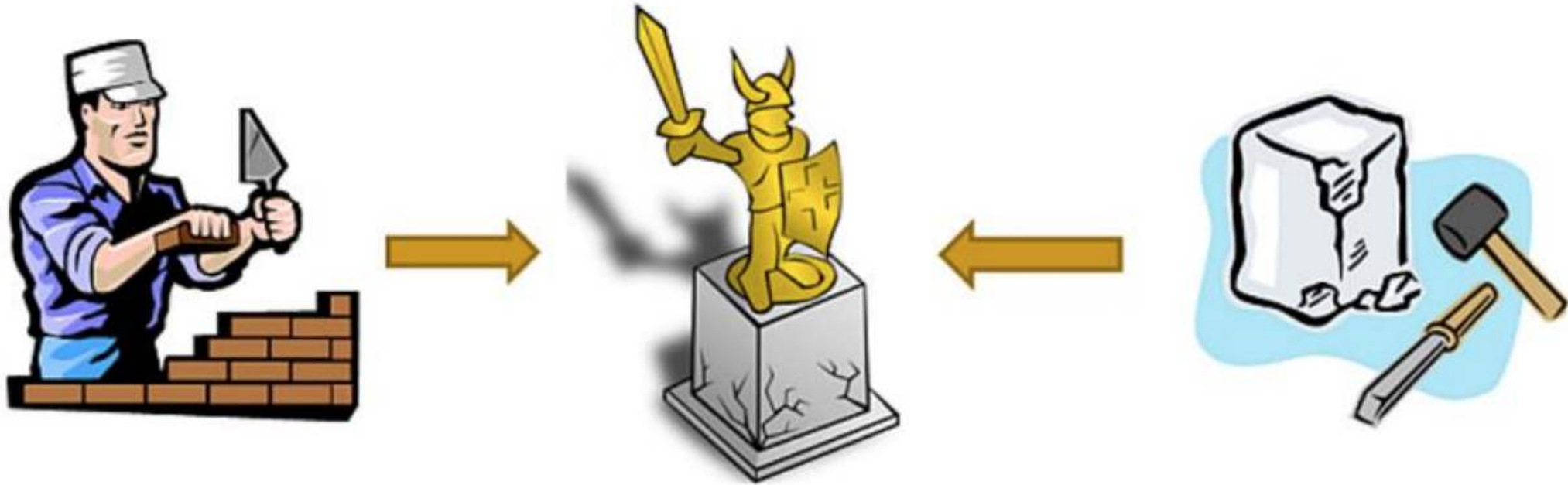
Teknik Fabrikasi Nanomaterial

Eka Nurfani

Program Studi Teknik Material, Institut Teknologi Sumatera

Manufacturing of Nanomaterials

Approaches: (1) Bottom-up; (2) Top-down

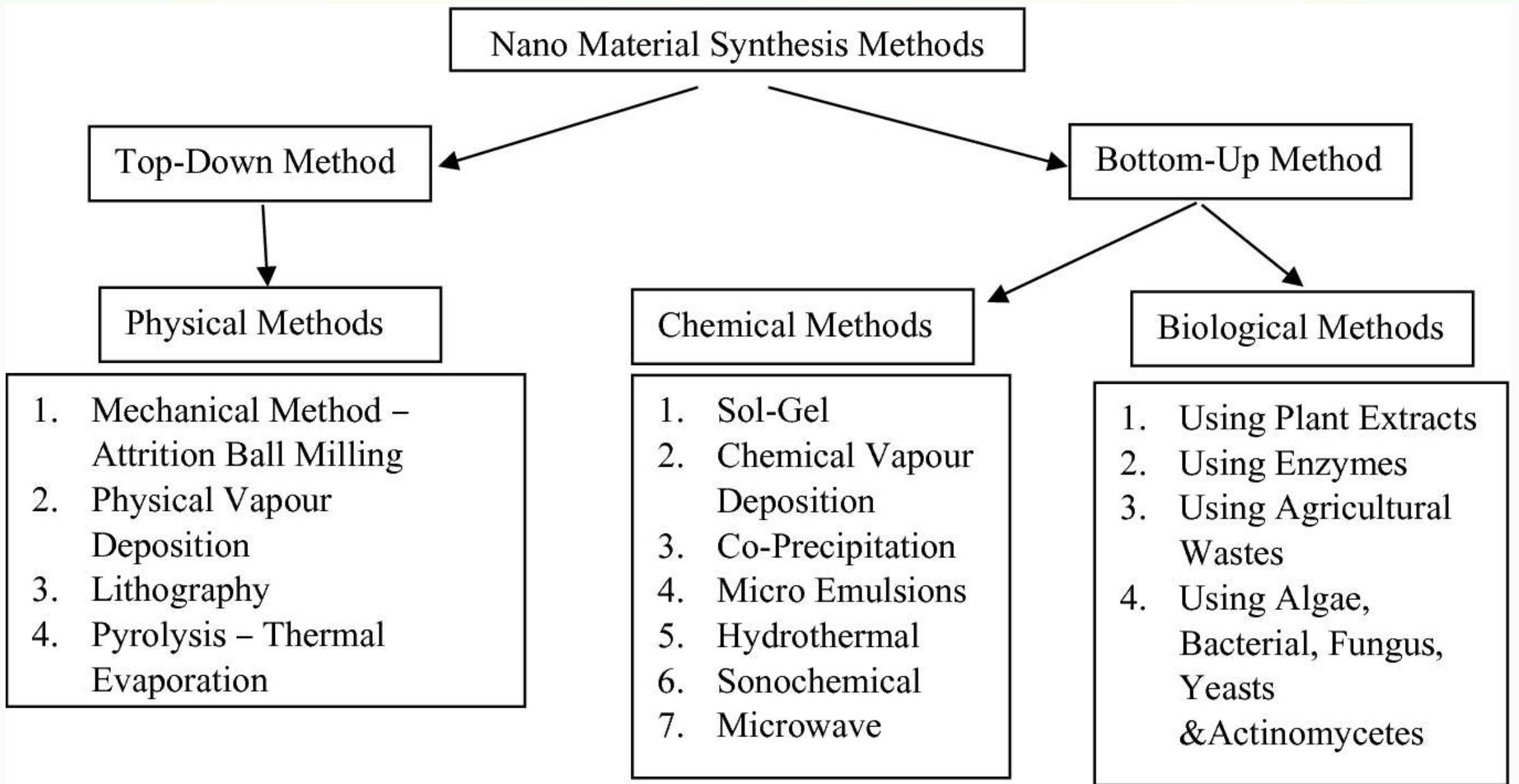


Bottom-up:

Building complex
structures from
building blocks

Top-down:

Fabrication of
Nanoscale structures
from the bulk material



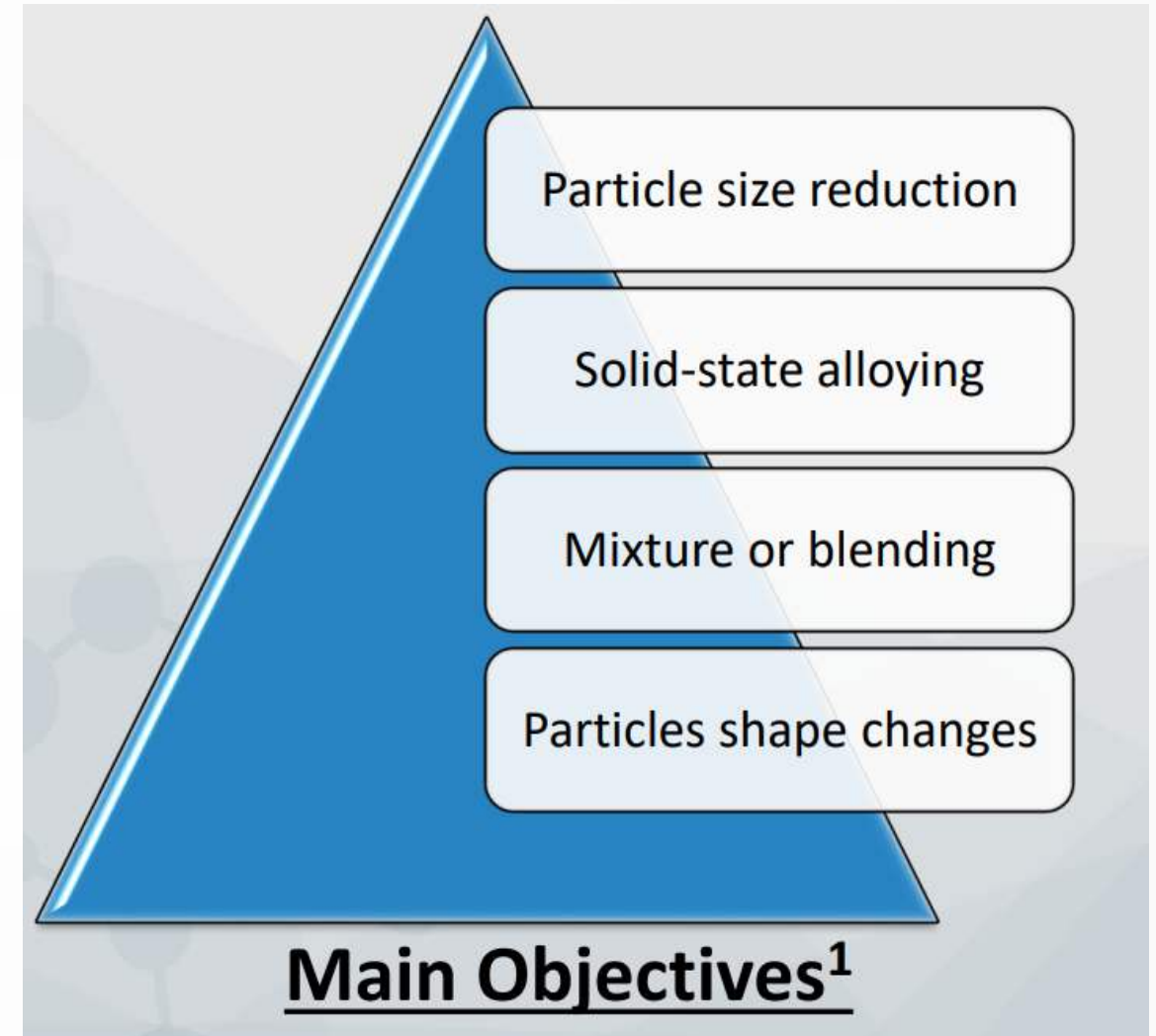
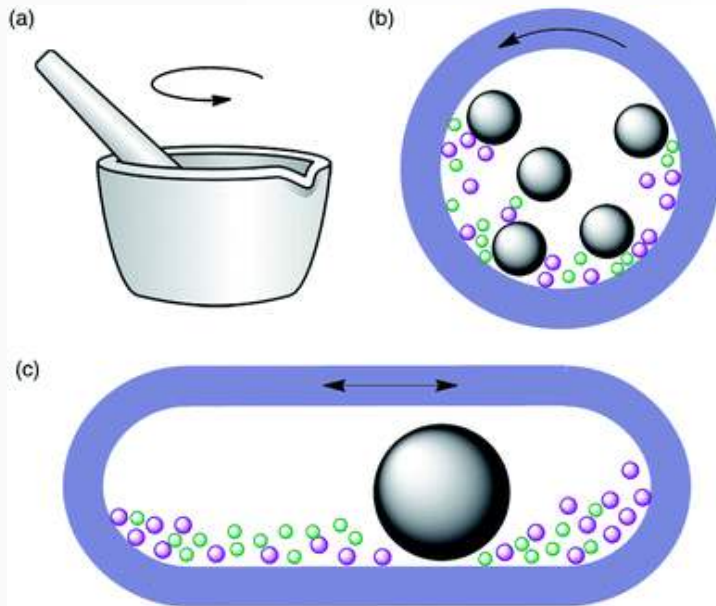
Physical Method

- **Mechanical ball milling**
- Physical Vapour Deposition
- Lithography
- Thermal Evaporation

Mechanical milling

Milling Applications:

- 1) Minerals Processing
- 2) Ceramics Processing
- 3) Powders Metallurgy



Type of Mill devices

- Types of mills for high-energy milling:

A) ball mill,

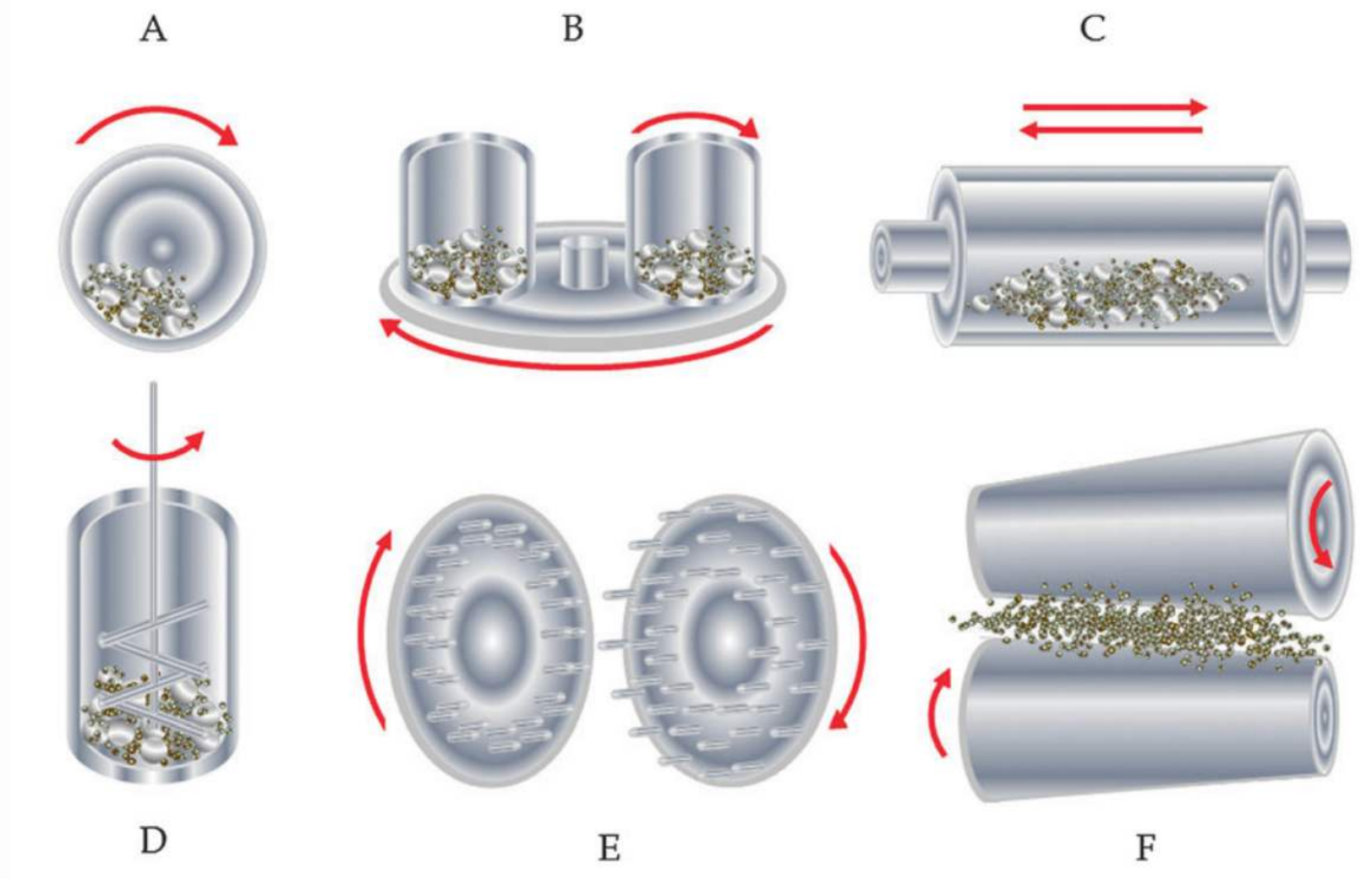
B) planetary mill,

C) vibration mill,

D) attritor (stirring ball mill),

E) pin mill,

F) rolling mill.



Mechanical milling

Tingkat penggilingan dipengaruhi oleh:

1. Waktu tinggal material di ruang penggilingan
2. Ukuran, kepadatan, dan jumlah bola
3. Sifat bola (kekerasan bahan penggilingan)
4. Kecepatan putaran silinder

Physical Method

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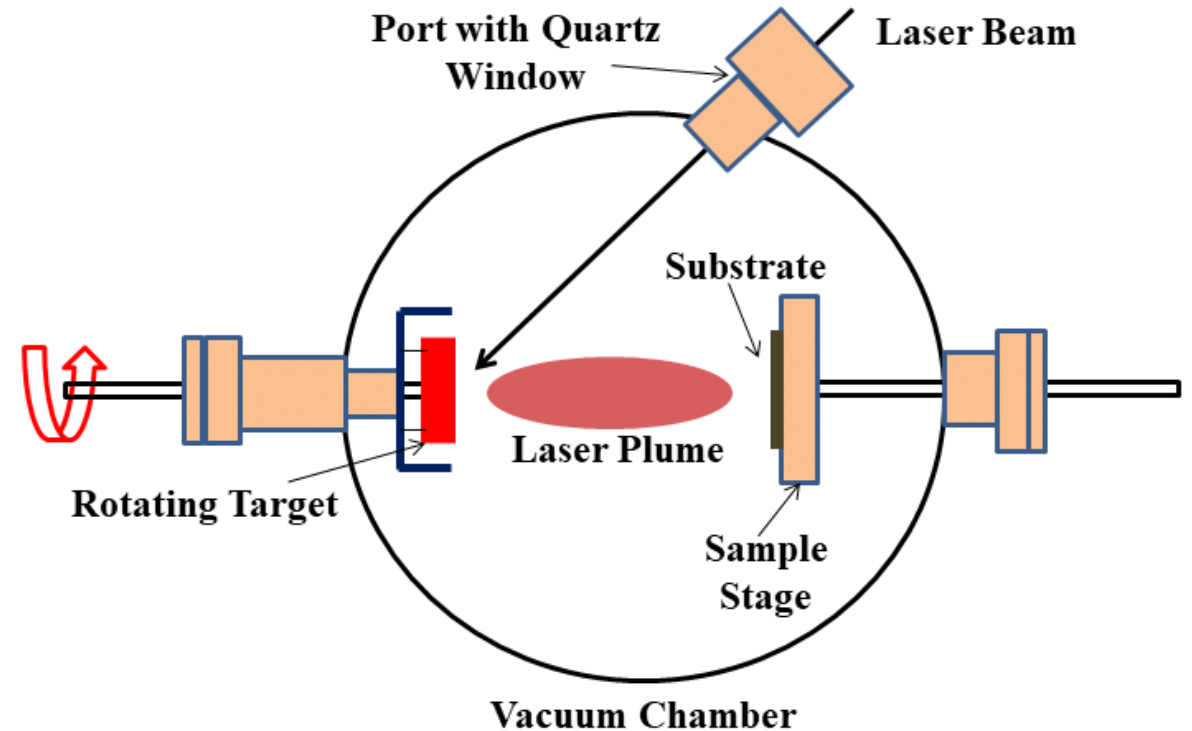
Physical Vapour Deposition

- 1) Pulsed laser deposition
- 2) Molecular beam epitaxy
- 3) Sputtering

Pulsed laser deposition

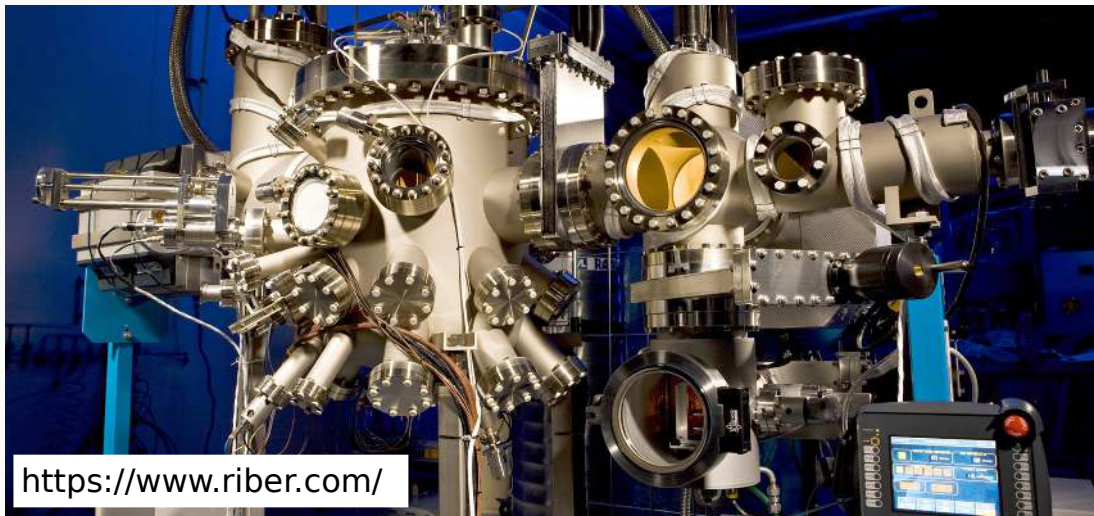
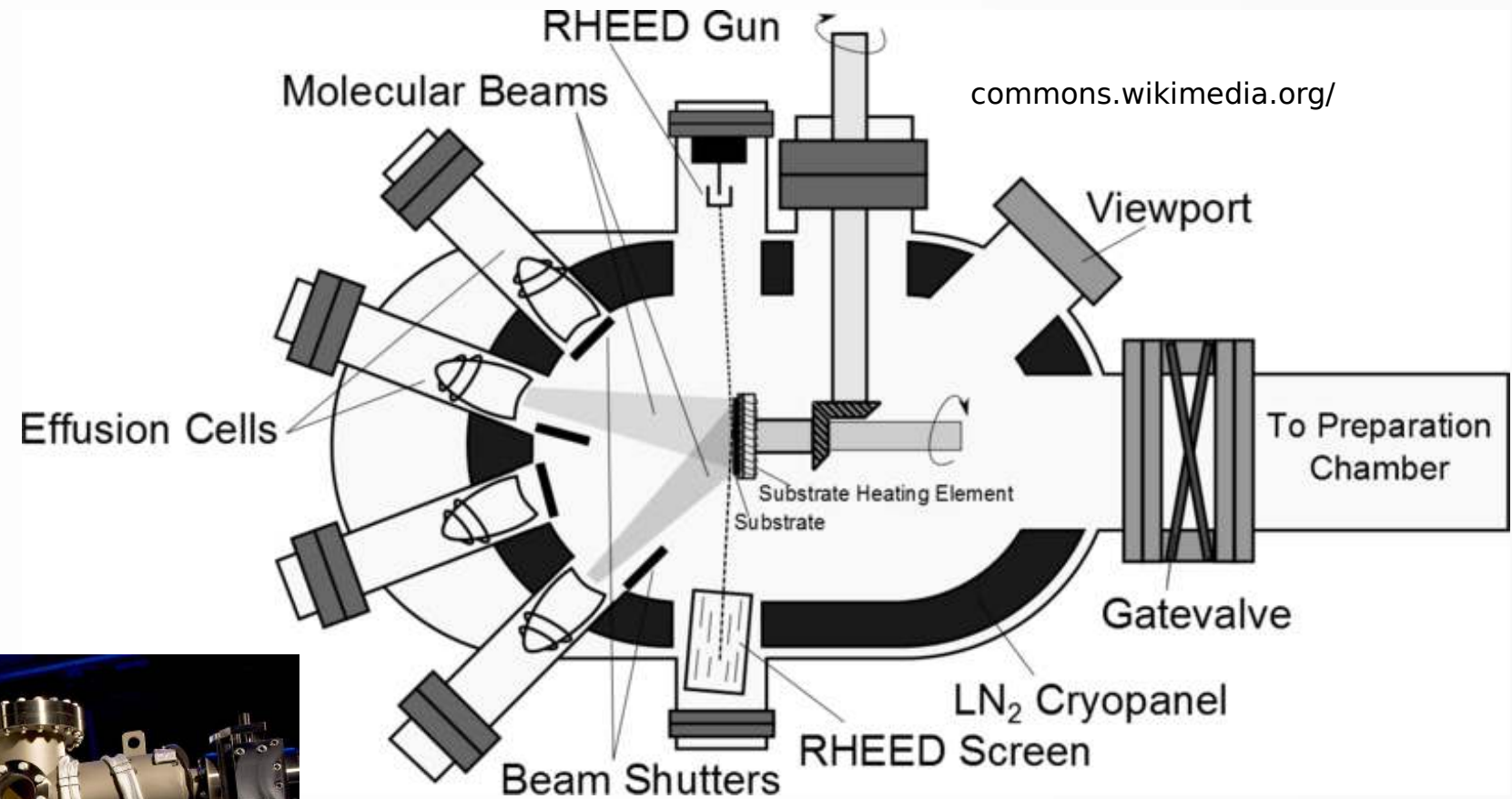


<http://www.pvdproducts.com/pulsed-laser-deposition-systems>



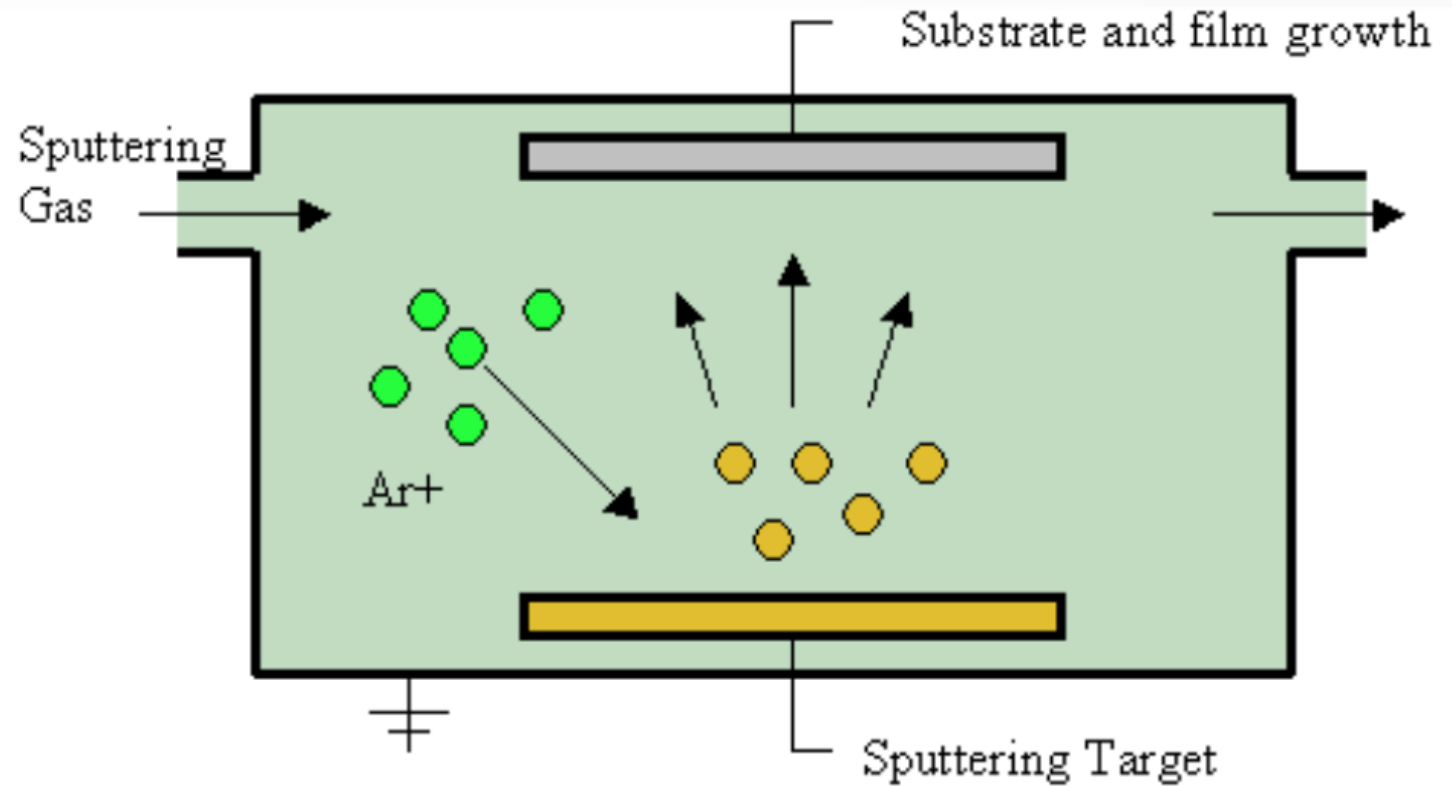
Pulsed laser deposition (PLD) adalah teknik deposisi uap fisik di mana sinar laser berdenyut daya tinggi difokuskan untuk menyerang target komposisi yang diinginkan. Bahan kemudian diuapkan dan diendapkan sebagai film tipis pada substrat yang menghadap target. Proses ini dapat terjadi dalam vakum ultra tinggi atau dengan adanya gas latar belakang, seperti oksigen ketika menyimpan film oksida.

Molecular beam epitaxy



Sputter deposition system

- Bombardment and removal of the cathode material with positive ions from a rare gas discharge.



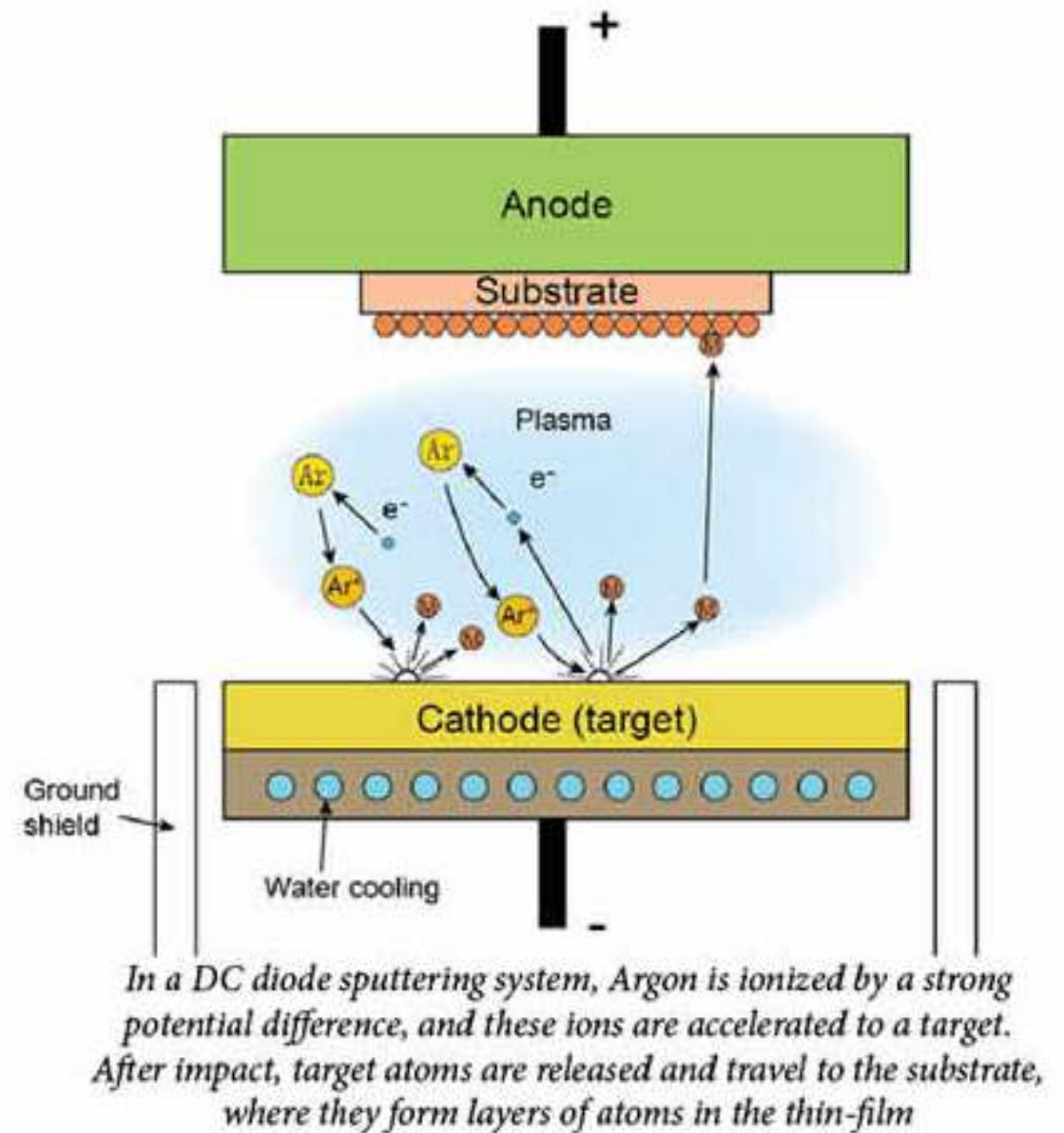
- | | |
|---------------------|--------------------|
| 1) vacuum chamber | 2) pumping system |
| 3) sputter sources | 4) vacuum gauges |
| 5) substrate holder | 6) Gas flow supply |

Mechanism of Sputtering

- Sputtering melibatkan tumbukan ion energik dengan permukaan target dan prosesnya biasanya mengarah pada ejeksi atom target.

TIPE:

- 1)DC (diode) sputtering
- 2)RF (radio frequency) sputtering
- 3)Magnetron sputtering
- 4)Reactive sputtering

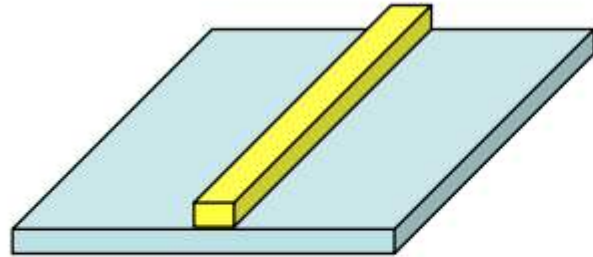


Physical Method

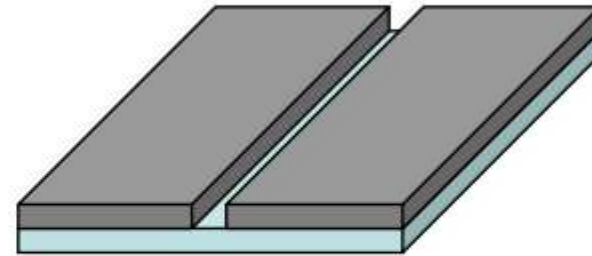
- Mechanical ball milling
- Physical Vapour Deposition
- **Lithography**
- Thermal Evaporation

Lithography: Basic concepts

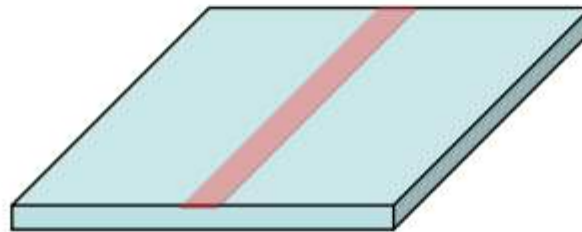
Some possible desired features



narrow line



narrow trench

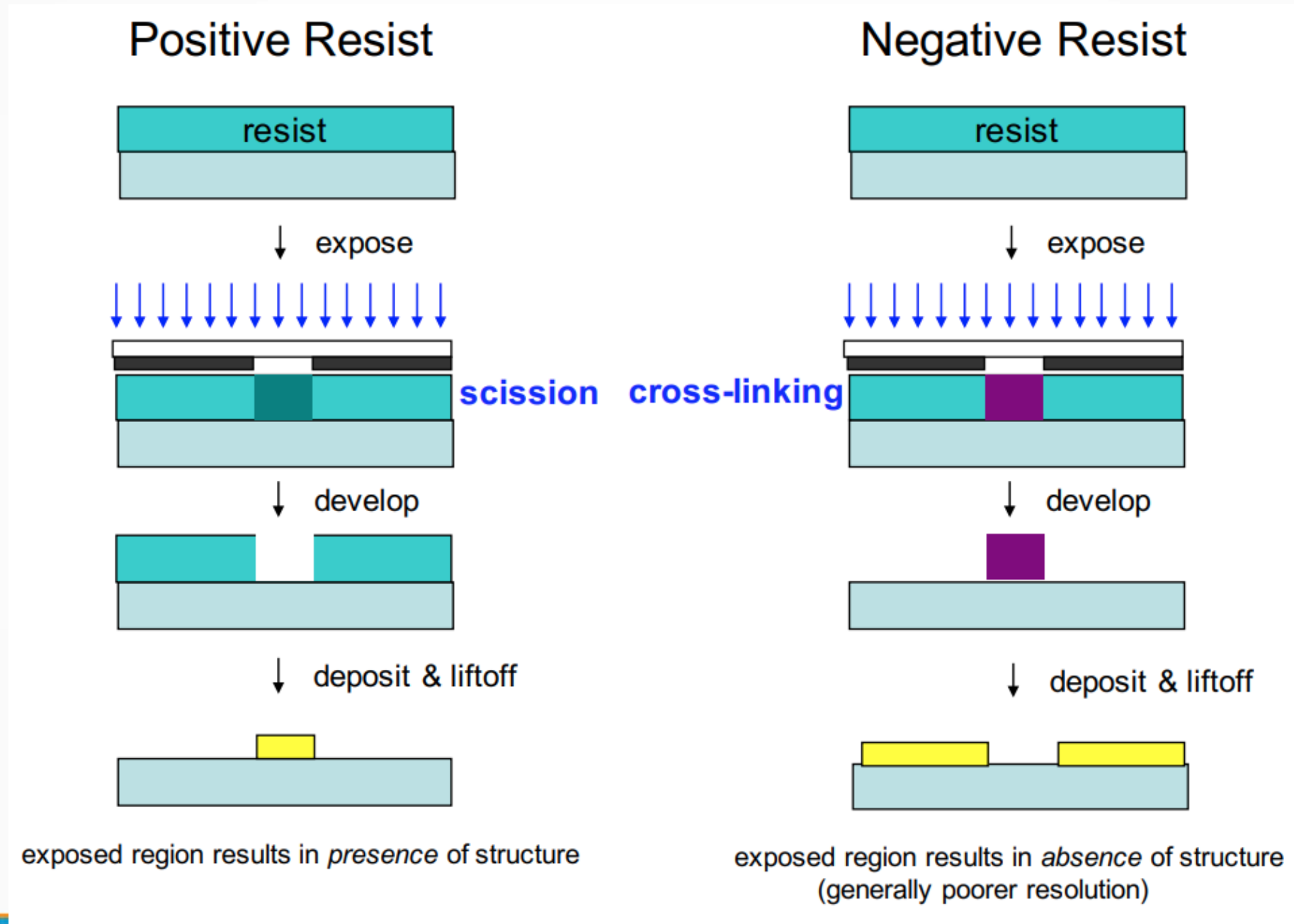


modified substrate

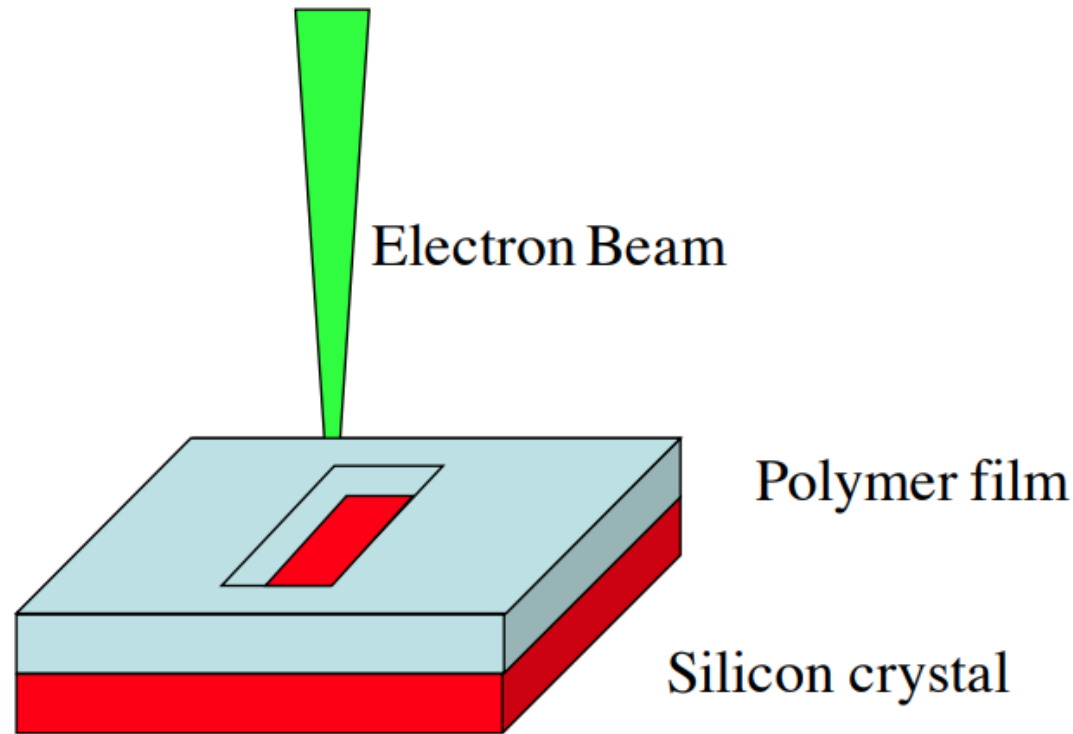
- **Photolithography**
- Electron-Beam Lithography
- X-ray Lithography
- Focused Ion-Beam Lithography
- Block Copolymer Lithography

- Nano Imprint Lithography
- Dip Pen Lithography
- Interference Lithography
- Contact Lithography
- EUV Lithography

Positive and Negative Resists

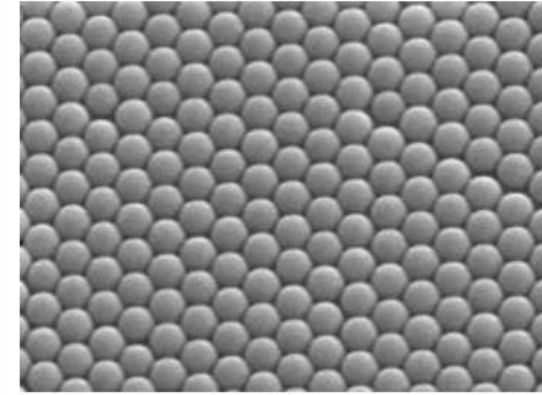


Electron-Beam Lithography

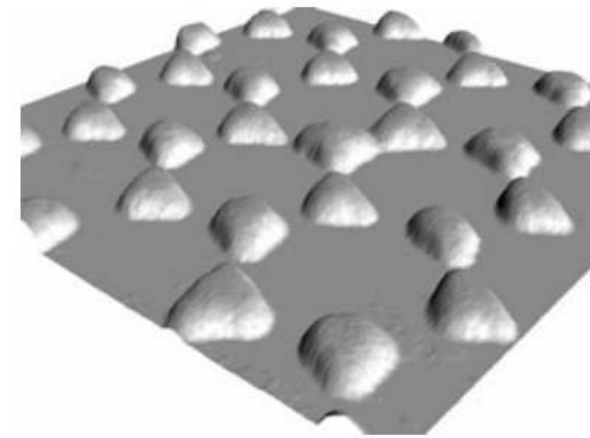


Nanoscopic Mask !

Down to 10 nm



itrc.org.tw



opticalproteomics.org

Physical Method

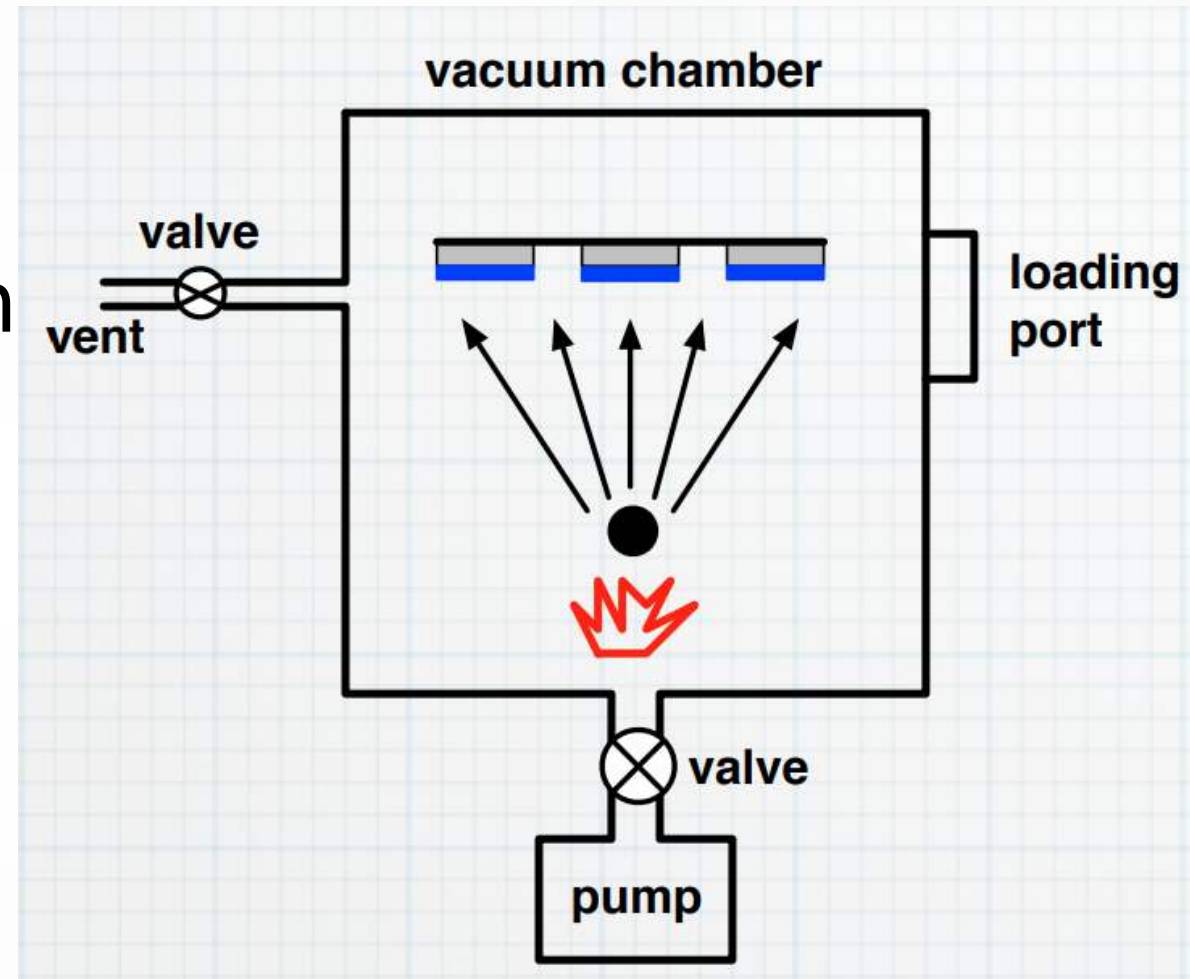
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- Physical Vapour Deposition
- Lithography
- **Thermal Evaporation**

Thermal evaporation

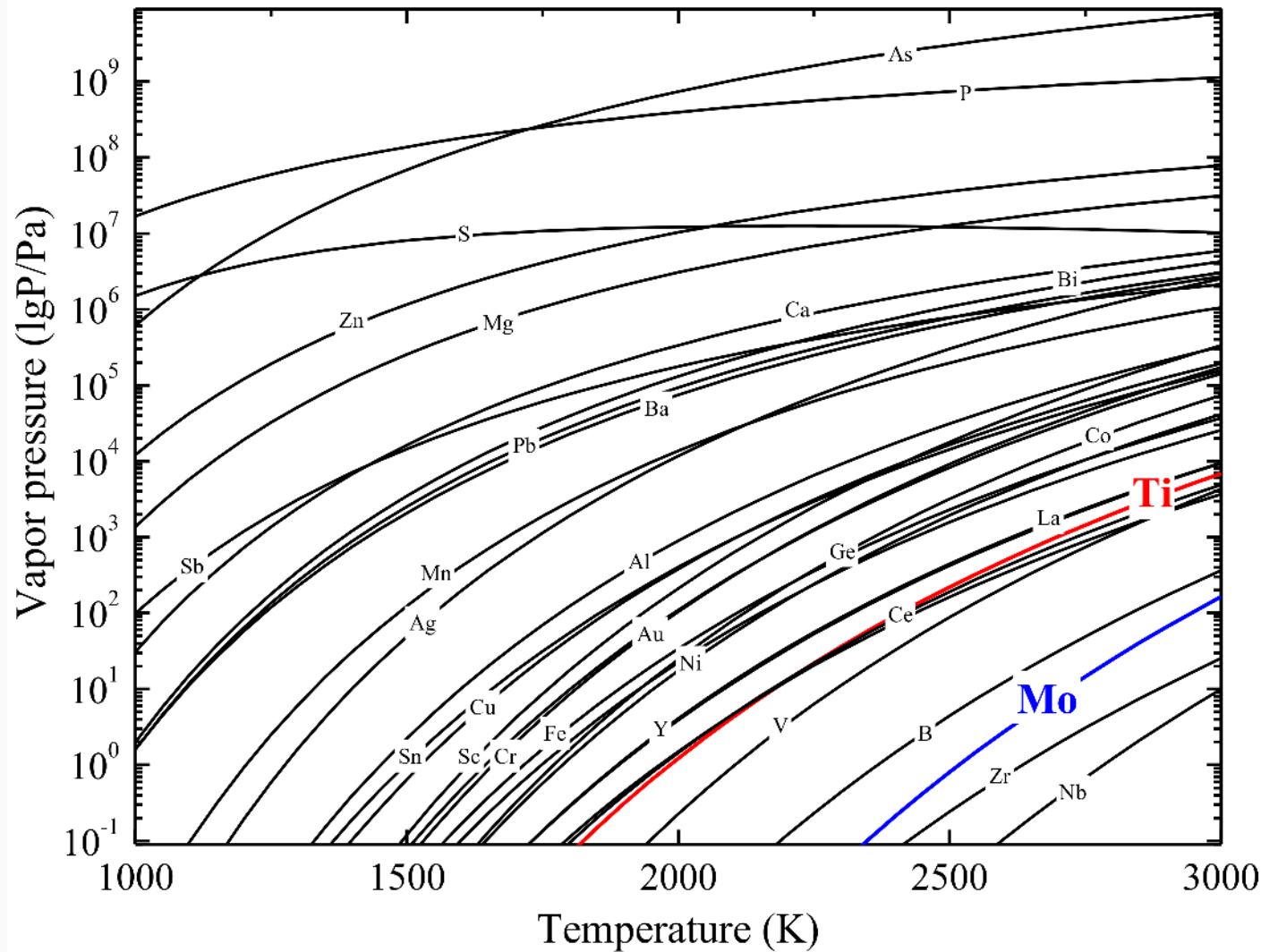
- Used to deposit thin layers (thin films) of metal on a substrate.
- Some metals films that are easily deposited by evaporation:
 - aluminum
 - chrome
 - gold
 - silver
 - titanium

Generic evaporation system

- Substrat menghadap ke bawah.
- Sumber logam yang akan diuapkan terletak di bawah wafer.
- Sumber dipanaskan ke titik di mana ia akan menguap.
- Uap mengembun pada substrat dingin (dan dinding ruang).



Vapor pressure curves



Agar dapat menguapkan suatu bahan secara efektif, bahan tersebut harus memiliki tekanan uap yang tinggi pada suhu yang wajar.

Heating the source

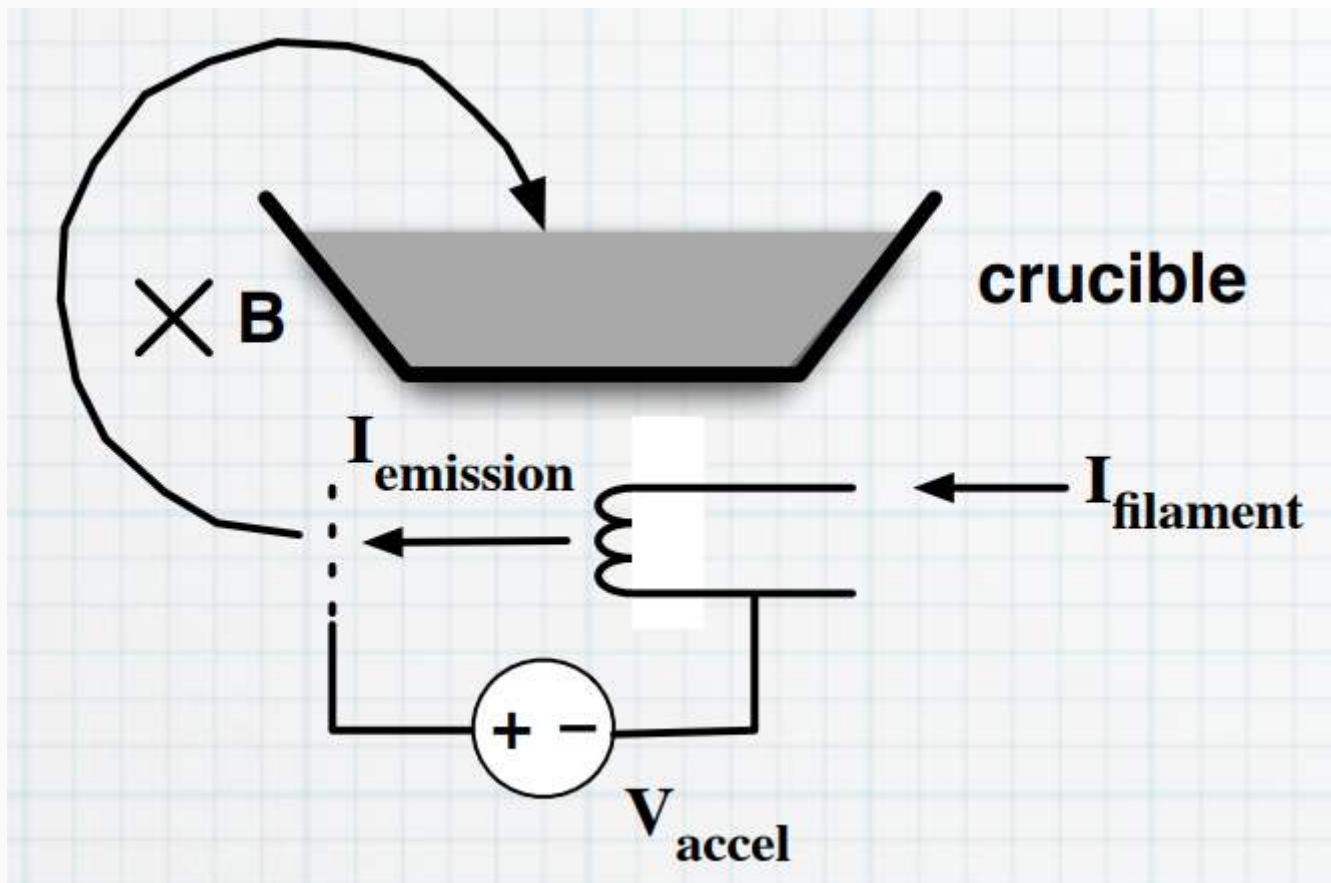
- Logam yang akan diuapkan dimasukkan ke dalam “keranjang” logam – biasanya terbuat dari Tantalum atau Tungsten.
- Arus besar melewati dudukan, memanaskannya secara resistif. Pada suhu yang cukup tinggi, bahan sumber akan mulai menguap.



Tungsten Boat

Electron-beam heating (e-beam evaporation)

- Bahan sumber ditempatkan dalam wadah (biasanya grafit).
- Sebuah filamen di bawah wadah dipanaskan dengan mengalirkan arus besar melaluinya ($\approx 100\text{ A}$)



- Beberapa elektron dalam filamen panas akan memiliki energi yang cukup untuk melepaskan diri (emisi termionik.)
- Elektron yang dipancarkan dipercepat dan kemudian dipelintir oleh medan magnet, membenturkannya ke bahan sumber, memanaskannya hingga suhu penguapan.

e-beam evaporation



TERIMA KASIH :)